The collision avoidance force that I am using follows a linear law with respect to the distance between 2 bots. It starts at 0 when two bots are at the sensing limit with respect to each other, and increases linearly as distance decreases.

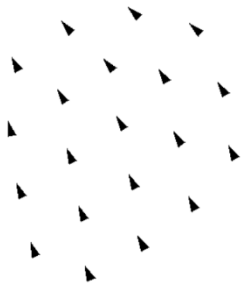
I will now specify a few scenarios with the scaling factor for each of the three forces, as well as the sensing radius:

1. k1 = 0.01

   k2 = 0.01

   k3 = 0.01

   r = 100



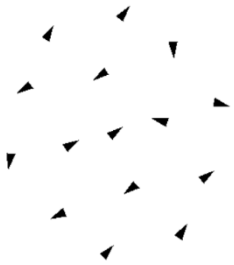
The boids move quite fast, and initially in quite a chaotic fashion. The system does eventually result in all boids forming a single flock. However, it takes a long amount of time to do so because fast moving boids sometimes fragment a larger flock, and because of the wrap-around at the boundaries, the forces acting on internal boids change when the outer boids pass through and come out at the other side of the screen. Also, sometimes when a faster moving smaller group comes close to a slower moving bigger group, the bigger group “poaches” some of the boids from the smaller group.

2. K1 = 0.005
   K2 = 0.01
   K3 = 0.01
   r = 200



The inter-boid spacing in a flock has increased as compared to the previous case. This scenario also has faster moving boids (because of the overall increase in collision avoidance forces), and thus it struggles to form a stable flock without its exterior members getting wrapped around. Overall, it looks more chaotic than the previous case.

3. K1 = 0.005
   K2 = 0.01
   K3 = 0.02
   r = 200

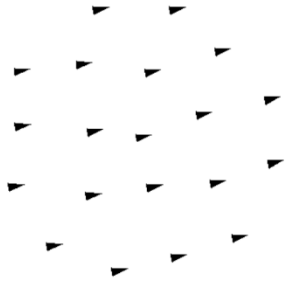




On increasing the scaling factor for the flock centering force, the system tends to form multiple clusters. As shown in the figure above, the system has no

problem forming stable clusters. However, it does struggle in the velocity alignment aspect. The larger flock in the picture above does not attain equilibrium velocity alignment easily. The velocities of the boids keep oscillating, with no net movement of the cluster.

4. K1 = 0.005
   K2 = 0.02
   K3 = 0.02
   r = 200



This system does attain a stable flock relatively easily compared to the above 3. Though the wrapping around mechanism does still cause problems, overall the system attains equilibrium considerably faster than the above three situations.